assignment_3_mta

October 27, 2025

1 Assignment 3

1.0.1 Due 9/23. Do four of five.

1.

- Open the NHANES (or Ames prices or college completion datasets, if you prefer)
- Find two categorical variables of interest (there are 198, and short descriptions are given in the `nhanes_meta_17_18.csv` file). Investigate their missing values (you don't have to focus on missing values for this analysis like we did with police use of force, but always be aware of how dirty the data are)
- Compute a contingency table for your categorical X and Y
- Discuss any interesting patterns (or lack of one) that you observe

[41]:
```python
import os
import pandas as pd
import seaborn as sns

os.chdir("/Users/mta/Desktop/Understanding Uncertainty/MTA4KD/data")

data = pd.read_csv("nhanes_data_17_18.csv")
meta = pd.read_csv("nhanes_meta_17_18.csv")

print("Participants:", data.shape[0])
print("Variables:", data.shape[1])
print("Metadata variables:", meta.shape[0])

display(data.head())
display(meta[['Variable', 'Label', 'Target']].head(200))
```

```
Participants: 8366
Variables: 198
Metadata variables: 197
```

```
/var/folders/_x/p9ds2_7j4cg3964ybwvy6rh00000gn/T/ipykernel_83926/2540604510.py:7
: DtypeWarning: Columns (142) have mixed types. Specify dtype option on import
or set low_memory=False.
  data = pd.read_csv("nhanes_data_17_18.csv")
```

```
   SEQN GeneralHealthCondition  EverBreastfedOrFedBreastmilk  \
```

```
0  93703.0                 NaN                           1.0
1  93704.0                 NaN                           1.0
2  93705.0                Good                           NaN
3  93706.0           Very good                           NaN
4  93707.0                Good                           NaN

   AgeStoppedBreastfeedingdays  AgeFirstFedFormuladays  \
0                        273.0                     1.0
1                         60.0                     3.0
2                          NaN                     NaN
3                          NaN                     NaN
4                          NaN                     NaN

   AgeStoppedReceivingFormuladays  AgeStartedOtherFoodbeverage  \
0                           365.0                        152.0
1                           365.0                        126.0
2                             NaN                          NaN
3                             NaN                          NaN
4                             NaN                          NaN

   AgeFirstFedMilkdays TypeOfMilkFirstFedWholeMilk  TypeOfMilkFirstFed2Milk  \
0                365.0       Whole or regular milk                      NaN
1                365.0       Whole or regular milk                      NaN
2                  NaN                         NaN                      NaN
3                  NaN                         NaN                      NaN
4                  NaN                         NaN                      NaN

   …  DaysSmokedCigsDuringPast30Days  AvgCigarettesdayDuringPast30Days  \
0  …                             NaN                               NaN
1  …                             NaN                               NaN
2  …                             NaN                               NaN
3  …                             NaN                               NaN
4  …                             NaN                               NaN

   TriedToQuitSmoking  TimesStoppedSmokingCigarettes  \
0                 NaN                            NaN
1                 NaN                            NaN
2                 NaN                            NaN
3                 NaN                            NaN
4                 NaN                            NaN

  HowLongWereYouAbleToStopSmoking UnitOfMeasureDayweekmonthyear_2_SMQ  \
0                             NaN                                 NaN
1                             NaN                                 NaN
2                             NaN                                 NaN
3                             NaN                                 NaN
4                             NaN                                 NaN
```

```
  CurrentSelfreportedHeightInches  CurrentSelfreportedWeightPounds  \
0                              NaN                              NaN
1                              NaN                              NaN
2                             63.0                            165.0
3                             68.0                            145.0
4                              NaN                              NaN

   TriedToLoseWeightInPastYear  TimesLost10LbsOrMoreToLoseWeight
0                          NaN                               NaN
1                          NaN                               NaN
2                          0.0                   11 times or more
3                          0.0                             Never
4                          NaN                               NaN

[5 rows x 198 columns]
    Variable                                     Label  \
0     HSD010                  General health condition
1     DBQ010          Ever breastfed or fed breastmilk
2     DBD030           Age stopped breastfeeding(days)
3     DBD041               Age first fed formula(days)
4     DBD050       Age stopped receiving formula(days)
..       …                                         …
192  SMQ852U     Unit of measure (day/week/month/year)
193   WHD010     Current self-reported height (inches)
194   WHD020     Current self-reported weight (pounds)
195   WHQ070         Tried to lose weight in past year
196   WHQ225  Times lost 10 lbs or more to lose weight

                                                Target
0     Both males and females 12 YEARS -\n\n\t\t\t15…
1     Both males and females 0 YEARS -\n\n\t\t\t6 Y…
2     Both males and females 0 YEARS -\n\n\t\t\t6 Y…
3     Both males and females 0 YEARS -\n\n\t\t\t6 Y…
4     Both males and females 0 YEARS -\n\n\t\t\t6 Y…
..                                                …
192   Both males and females 18 YEARS -\n\n\t\t\t15…
193   Both males and females 16 YEARS -\n\n\t\t\t15…
194   Both males and females 16 YEARS -\n\n\t\t\t15…
195   Both males and females 16 YEARS -\n\n\t\t\t15…
196   Both males and females 16 YEARS -\n\n\t\t\t15…

[197 rows x 3 columns]
```

I'm choosing to analyze 1) Race/Hispanic origin w/ NH Asian (RIDRETH3) and 2) Doctor ever said you had arthritis (MCQ160A). These are both categorical variables.

[42]:
```
print([col for col in data.columns if "Race" in col])
print([col for col in data.columns if "Arth" in col or "arth" in col])
```

```
['RacehispanicOrigin', 'RacehispanicOriginWNhAsian']
['DoctorEverSaidYouHadArthritis']
```

[43]:
```
print(meta.columns.tolist())
```

```
['VariableNameLong', 'Variable', 'Type', 'Format', 'Informat', 'Label',
'Source', 'VariableName', 'SASLabel', 'EnglishText', 'Target',
'VariableNameLong_variable_df', 'Source_variable_df', 'EnglishInstructions',
'HardEdits', 'Recoded', 'CustomRecoding', 'nNonNA']
```

[44]:
```
col_race = "RacehispanicOriginWNhAsian"
col_arthritis = "DoctorEverSaidYouHadArthritis"

for var_id, col in [("RIDRETH3", col_race), ("MCQ160A", col_arthritis)]:
    missing_pct = data[col].isna().mean() * 100
    print(f"{col} ({var_id}) - Missing: {missing_pct:.2f}%")

def summarize_categories(var_id, col_name):
    meta_info = meta[meta["Variable"] == var_id][["Variable", "EnglishText",␣
↪"Label"]]
    print(f"\n--- {col_name} ({var_id}) ---")
    print(meta_info.to_string(index=False))

    counts = data[col_name].value_counts(dropna=False).reset_index()
    counts.columns = [col_name, "Frequency"]
    print(counts)

summarize_categories("RIDRETH3", col_race)
summarize_categories("MCQ160A", col_arthritis)
```

```
RacehispanicOriginWNhAsian (RIDRETH3) - Missing: 0.00%
DoctorEverSaidYouHadArthritis (MCQ160A) - Missing: 37.25%

--- RacehispanicOriginWNhAsian (RIDRETH3) ---
Variable
EnglishText                            Label
RIDRETH3 Recode of reported race and Hispanic origin information, with Non-
Hispanic Asian Category Race/Hispanic origin w/ NH Asian
            RacehispanicOriginWNhAsian  Frequency
0                   Non-Hispanic White       2792
1                   Non-Hispanic Black       1949
2                     Mexican American       1229
3                   Non-Hispanic Asian       1083
4                       Other Hispanic        738
5  Other Race - Including Multi-Racial        575
```

```
--- DoctorEverSaidYouHadArthritis (MCQ160A) ---
Variable
EnglishText                                   Label
 MCQ160A Has a doctor or other health professional ever told {you/SP} that
{you/s/he} . . .had arthritis (ar-thry-tis)? Doctor ever said you had arthritis
   DoctorEverSaidYouHadArthritis  Frequency
0                            0.0       3645
1                            NaN       3116
2                            1.0       1605
```

[45]:

```python
selected_vars = [col_race, col_arthritis]

missing_counts = pd.DataFrame({
    var: data[var].isna().value_counts()
    for var in selected_vars
}).T

missing_pct = pd.DataFrame({
    var: data[var].isna().value_counts(normalize=True) * 100
    for var in selected_vars
}).T

print("\nMissingness Counts")
print(missing_counts)

print("\nMissingness Percentages")
print(missing_pct.round(2))
```

```
Missingness Counts
                                False   True
RacehispanicOriginWNhAsian     8366.0     NaN
DoctorEverSaidYouHadArthritis  5250.0  3116.0

Missingness Percentages
                                False  True
RacehispanicOriginWNhAsian     100.00    NaN
DoctorEverSaidYouHadArthritis   62.75  37.25
```

[46]:

```python
contingency_counts = pd.crosstab(
    data[col_race],
    data[col_arthritis],
    margins=True,
    dropna=False
)

contingency_row_pct = pd.crosstab(
    data[col_race],
```

```python
    data[col_arthritis],
    normalize="index",
    dropna=False
) * 100

contingency_col_pct = pd.crosstab(
    data[col_race],
    data[col_arthritis],
    normalize="columns",
    dropna=False
) * 100

print("\nContingency Table: Counts")
print(contingency_counts)

print("\nContingency Table: Row Percentages (within each race)")
print(contingency_row_pct.round(2))

print("\nContingency Table: Column Percentages (within arthritis response)")
print(contingency_col_pct.round(2))
```

```
Contingency Table: Counts
DoctorEverSaidYouHadArthritis          0.0   1.0  NaN   All
RacehispanicOriginWNhAsian
Mexican American                       544   152  533  1229
Non-Hispanic Asian                     636   121  326  1083
Non-Hispanic Black                     841   395  713  1949
Non-Hispanic White                    1097   707  988  2792
Other Hispanic                         359   133  246   738
Other Race - Including Multi-Racial    168    97  310   575
All                                   3645  1605    0  8366

Contingency Table: Row Percentages (within each race)
DoctorEverSaidYouHadArthritis           0.0    1.0    NaN
RacehispanicOriginWNhAsian
Mexican American                      44.26  12.37  43.37
Non-Hispanic Asian                    58.73  11.17  30.10
Non-Hispanic Black                    43.15  20.27  36.58
Non-Hispanic White                    39.29  25.32  35.39
Other Hispanic                        48.64  18.02  33.33
Other Race - Including Multi-Racial   29.22  16.87  53.91

Contingency Table: Column Percentages (within arthritis response)
DoctorEverSaidYouHadArthritis           0.0    1.0    NaN
RacehispanicOriginWNhAsian
Mexican American                      14.92   9.47  17.11
Non-Hispanic Asian                    17.45   7.54  10.46
```

```
Non-Hispanic Black                     23.07  24.61  22.88
Non-Hispanic White                     30.10  44.05  31.71
Other Hispanic                          9.85   8.29   7.89
Other Race - Including Multi-Racial     4.61   6.04   9.95
```

Looking at the contingency table, two things jump out. First, the row percentages: within each race, Non-Hispanic Whites have the highest reported arthritis rate (about 25%), followed by Non-Hispanic Blacks (20%). Mexican Americans and Asians are lower, closer to 12%. So prevalence clearly isn't even across groups. Second, the column percentages: when you look at the arthritis "Yes" pool, it's dominated by Whites (44%) and Blacks (25%), meaning those two groups make up the bulk of reported arthritis cases in this dataset. The big caveat is missingness — it's huge, 30–50% depending on the group — so you can't treat these as clean prevalence estimates. Still, the directional pattern is obvious: arthritis is more common among Whites and Blacks, less so among Mexican Americans and Asians.

2.

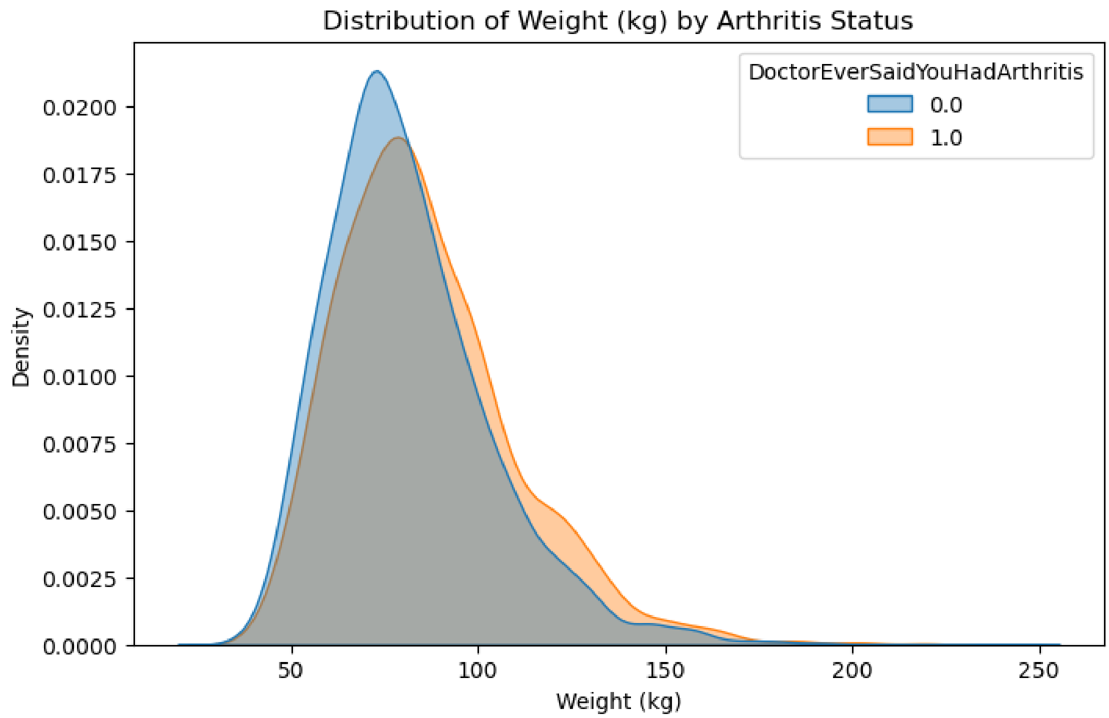
- Open the NHANES dataset
- Find a categorical and numeric variable of interest (there are 198, and short descriptions are given in the `nhanes_meta_17_18.csv` file). Investigate their missing values (you don't have to focus on missing values for this analysis, but always be aware of them)
- Make descriptive tables and grouped kernel density plots to represent the variation in your numeric Y conditional on your categorical X
- Discuss any interesting patterns (or lack of one) that you observe

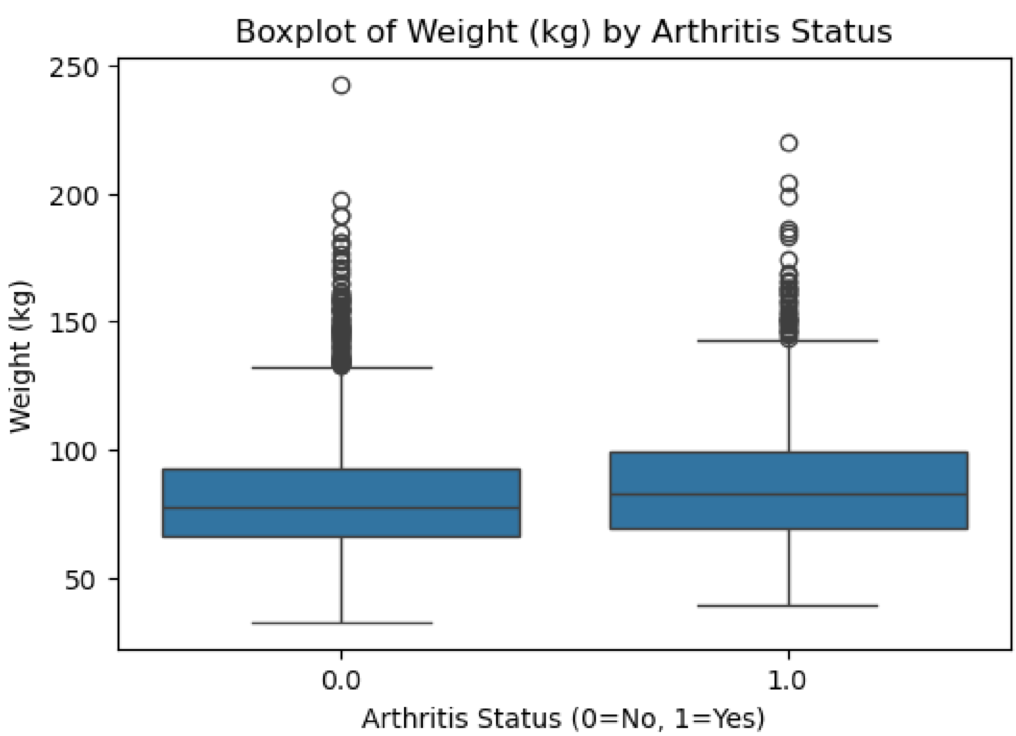
I'm choosing to analyze 1) Weight KG (BMXWT) and 2) Doctor ever said you had arthritis (MCQ160A).

[47]:
```python
col_cat = "DoctorEverSaidYouHadArthritis"
col_num = "WeightKg"



for col in [col_cat, col_num]:
    missing_pct = data[col].isna().mean() * 100
    print(f"{col} - Missing: {missing_pct:.2f}%")

desc_stats = data.groupby(col_cat)[col_num].describe()
print("\nDescriptive Statistics (Weight by Arthritis status):")
print(desc_stats)

plt.figure(figsize=(8,5))
sns.kdeplot(
    data=data,
    x=col_num,
    hue=col_cat,
    common_norm=False,
    fill=True,
    alpha=0.4
)
```

```
plt.title("Distribution of Weight (kg) by Arthritis Status")
plt.xlabel("Weight (kg)")
plt.ylabel("Density")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(
    data=data,
    x=col_cat,
    y=col_num
)
plt.title("Boxplot of Weight (kg) by Arthritis Status")
plt.xlabel("Arthritis Status (0=No, 1=Yes)")
plt.ylabel("Weight (kg)")
plt.show()
```

```
DoctorEverSaidYouHadArthritis - Missing: 37.25%
WeightKg - Missing: 1.48%

Descriptive Statistics (Weight by Arthritis status):
                               count       mean        std   min   25%   50%  \
DoctorEverSaidYouHadArthritis
0.0                           3600.0  81.367611  22.227085  32.6  66.1  77.7
1.0                           1573.0  86.150223  23.990346  39.3  69.3  82.6

                                75%    max
DoctorEverSaidYouHadArthritis
0.0                           92.7  242.6
1.0                           98.8  219.6
```


Distribution of Weight (kg) by Arthritis Status
DoctorEverSaidYouHadArthritis
0.0
1.0
Density
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125
0.0150
0.0175
0.0200
50
100
150
200
250
Weight (kg)


Boxplot of Weight (kg) by Arthritis Status
Weight (kg)
50
100
150
200
250
0.0
1.0
Arthritis Status (0=No, 1=Yes)

Looking at the stats and plots, there's a clear weight gap between people who reported arthritis and those who didn't. Average weight is about 86 kg in the arthritis group vs 81 kg in the no-arthritis group. The kernel density plot shows the arthritis group's distribution shifted slightly to the right — heavier on average, with a longer right tail. The boxplot tells the same story: higher median and more extreme outliers among those with arthritis. Missingness is low here (~1.5% for weight, ~37% for arthritis status), so this pattern isn't just noise. The takeaway is straightforward: in this sample, higher weight tracks with higher arthritis prevalence.

3. We showed that the mean and median could be discovered by minimizing various kinds of loss functions; this is what machine learning is. To make a prediction $\hat{y}(z)$ of Y when $X = z$, minimize the mean squared error:

$$MSE(\hat{y}(z)) = \frac{1}{N}\sum_{i=1}^{N} \{y_i - \hat{y}(z)\}^2 \frac{1}{h}k\left(\frac{z - x_i}{h}\right)$$

   Show that the solution to this problem is the LCLS/Naradaya-Watson estimator.

$$MSE(\hat{y}(z)) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}(z))^2\, w_i(z), \quad w_i(z) = \frac{1}{h}k\Big(\frac{z - x_i}{h}\Big).$$

Step 1: Differentiate wrt $\hat{y}(z)$

$$\frac{\partial}{\partial \hat{y}(z)} MSE(\hat{y}(z)) = \frac{1}{N}\sum_{i=1}^{N} 2(\hat{y}(z) - y_i)w_i(z) = \frac{2}{N}\left[\hat{y}(z)\sum_{i=1}^{N} w_i(z) - \sum_{i=1}^{N} w_i(z)y_i\right].$$

Step 2: Set equal to zero and solve

$$\hat{y}(z)\sum_{i=1}^{N} w_i(z) = \sum_{i=1}^{N} w_i(z)y_i,$$

$$\hat{y}(z) = \frac{\sum_{i=1}^{N} w_i(z)y_i}{\sum_{i=1}^{N} w_i(z)}.$$

Step 3: Plug back in kernel weights

Since

$$w_i(z) = \frac{1}{h}k\Big(\frac{z - x_i}{h}\Big),$$

we get:

$$\hat{y}(z) = \frac{\sum_{i=1}^{N} k\big(\frac{z - x_i}{h}\big)\, y_i}{\sum_{i=1}^{N} k\big(\frac{z - x_i}{h}\big)}.$$

This is the **Nadaraya–Watson estimator** - a weighted average of the y_i's, with weights coming from the kernel.

- Small h: sharp, twitchy, low bias, high variance. Like trading tick-by-tick — overfit and volatile.

- Large h: smooth, flat, high bias, low variance. Like averaging across decades — stable, but you miss the details.

The *estimator* is just balancing how much noise we want to eat versus how much structure we're willing to miss.

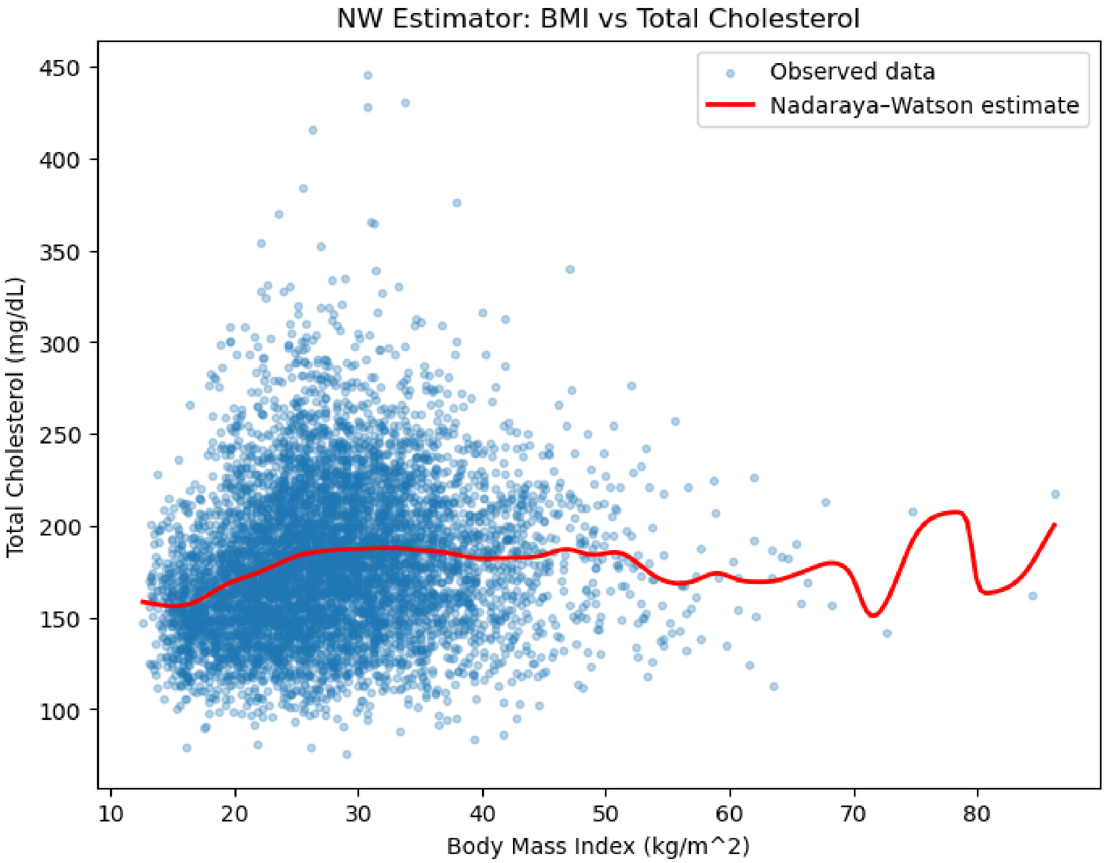
4.

- Write a class or set of functions that implement the LCLS/Naradaya-Watson estimator, using the Silverman plug-in estimate for the conditioning variable X as the bandwidth.
- From one of the course data sets, find two numeric variables of interest, analyze their relationship with the the LCLS/Naradaya-Watson estimator, and discuss your results.

[48]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm

col_x = "BodyMassIndexKgm2"
col_y = "TotalCholesterolMgdl"

subset = data[[col_x, col_y]].dropna()
X = subset[col_x].values
Y = subset[col_y].values



n = len(X)
h = 1.06 * np.std(X) * n ** (-1/5)

def nw_estimator(x0, X, Y, h):
    weights = norm.pdf((x0 - X) / h)
    return np.sum(weights * Y) / np.sum(weights)

x_grid = np.linspace(X.min(), X.max(), 200)
y_hat = np.array([nw_estimator(x0, X, Y, h) for x0 in x_grid])

plt.figure(figsize=(8,6))
plt.scatter(X, Y, s=10, alpha=0.3, label="Observed data")
plt.plot(x_grid, y_hat, color="red", lw=2, label="Nadaraya-Watson estimate")
plt.xlabel("Body Mass Index (kg/m^2)")
plt.ylabel("Total Cholesterol (mg/dL)")
plt.title("NW Estimator: BMI vs Total Cholesterol")
plt.legend()
```

```python
plt.show()

print("Descriptive stats:")
print(subset.describe())
```




```
Descriptive stats:
       BodyMassIndexKgm2  TotalCholesterolMgdl
count        6641.000000           6641.000000
mean           27.833444            179.982232
std             7.913572             40.647648
min            12.600000             76.000000
25%            22.300000            150.000000
50%            26.900000            176.000000
75%            32.100000            204.000000
max            86.200000            446.000000
```

The scatterplot shows the relationship between BMI and Total Cholesterol. The raw data are highly variable, but the Nadaraya–Watson estimator provides a smoothed curve that highlights the average trend. Cholesterol levels tend to increase slightly as BMI rises from low to mid-range, then flatten, with some increase again at higher BMI values.

The descriptive statistics confirm the wide spread in both variables: BMI (std 7.9) and cholesterol (std 49.6). Missingness is relatively modest for BMI (4%) but higher for cholesterol (19%).

Overall, the estimator suggests a weak positive association between BMI and cholesterol, though with considerable noise. The pattern indicates that individuals with higher BMI often show higher cholesterol on average, but the relationship is far from deterministic, and there is substantial variation across the range of BMI values.

5.

- In any of the available data sets, investigate the relationships between pairs of variables (X, Y) with a scatterplot and CEF (for example, price on area)
- Is this relationship plausibly causal, or are there missing variables that might explain at least part of the relationship between your variables? These can be "conceptual" rather than "practical"; for example, 'talent' or 'grit' probably explain education outcomes, but are almost impossible to measure. We are asking whether there are hypothetical **threats to causal identification** of the effect of X on Y.
- Explain how, regardless of the threat to causal identification, you can still use your model to predict Y given X, as long as you don't intervene in the system to control the outcome